F - Second Gap

分值：525 分

Problem Statement

You are given an integer N and an integer sequence $D=(D_1,D_2,\ldots,D_{N-1})$ of length $N-1$.

给定一个整数 N 和一个长度为 $N-1$ 的整数序列 $D=(D_1,D_2,\ldots,D_{N-1})$。

Find the number, modulo 998244353, of permutations $P=(P_1,P_2,\ldots,P_N)$ of $(1,2,\ldots,N)$ that satisfy the following condition.

求满足以下条件的 $(1,2,\ldots,N)$ 的排列 $P=(P_1,P_2,\ldots,P_N)$ 的个数，答案对 998244353 取模。

- For each $1\leq i\leq N-1$, let P_a and P_b be the elements with the largest and the second largest values, respectively, among $(P_i,P_{i+1},\ldots,P_N)$; then $|a-b|=D_i$.
- 对于每个 $1\leq i\leq N-1$，设 P_a 和 P_b 分别是 $(P_i,P_{i+1},\ldots,P_N)$ 中值最大和第二大的元素，则满足 $|a-b|=D_i$。

Constraints

- $2\leq N\leq 2\times 10^5$
- $1\leq D_i\leq N-i$
- All input values are integers.
- $2\leq N\leq 2\times 10^5$
- $1\leq D_i\leq N-i$
- 所有输入值均为整数。

Input

The input is given from Standard Input in the following format:

输入按照以下格式从标准输入给出：

N
$D_1\ D_2\ \ldots\ D_{N-1}$

Output

Output the number, modulo 998244353, of permutations satisfying the condition.

输出满足条件的排列的个数，答案对 998244353 取模。

Sample Input 1

```
3
1 1
```

Sample Output 1

```
4
```

For example, we can verify that $(2, 3, 1)$ satisfies the condition as follows.

例如，我们可以按如下方式验证 $(2, 3, 1)$ 满足条件：

- We have $(P_1, P_2, P_3) = (2, 3, 1)$. The largest value is P_2 and the second largest value is P_1, and $|2 - 1| = 1 = D_1$.
- 我们有 $(P_1, P_2, P_3) = (2, 3, 1)$。最大值为 P_2，第二大值为 P_1，且 $|2 - 1| = 1 = D_1$。

- We have $(P_2, P_3) = (3, 1)$. The largest value is P_2 and the second largest value is P_3, and $|2 - 3| = 1 = D_2$.
- 我们有 $(P_2, P_3) = (3, 1)$。最大值为 P_2，第二大值为 P_3，且 $|2 - 3| = 1 = D_2$。

Four permutations satisfy the condition: $(1, 2, 3), (1, 3, 2), (2, 3, 1), (3, 2, 1)$. Thus, output 4.

共有四个排列满足条件：$(1, 2, 3), (1, 3, 2), (2, 3, 1), (3, 2, 1)$。因此输出 4。

Sample Input 2

```
5
1 2 2 1
```

Sample Output 2

```
0
```

Sample Input 3

```
15
4 4 4 4 4 4 3 2 2 2 2 2 1 1
```

Sample Output 3

```
70270200
```